

USB-CAN Bus Adapter (CANalyst-II Analyzer)

Out-of-the-Box Test Instructions (With Fault Tolerant CAN)

Instruction Manual Version: V2.04

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I. Introduction

First of all, thank you for purchasing and using our products, we will always be happy to serve you!

This document is intended as a guide for first-time buyers of CANalyst-II analyzers (top of the line version with fault-tolerant CAN) to test whether the CANalyst-II analyzer they have received works properly.

It can be ignored by experienced users or users who are familiar with the operation of the CANalyst-II analyzer.

II. Test Steps

When you receive the CANalyst-II analyzer, please follow the steps described below in order to test it.

1. Install the Driver for the Hardware

Connect the CANalyst-II analyzer to one of the USB ports of your computer with the standard accessory "USB cable", and the computer will pop up the "Driver Installation Wizard".

Please refer to *"3. USB Driver Installation and Uninstallation Manual.pdf"* for detailed steps on driver installation.

2. Installation of Test Software

For detailed step-by-step instructions on the installation of the test software, please refer to *"4. USB-CAN Tool Debugging Software Installation and Usage Manual.pdf"*.

3. Connecting the Two Channels of the USB-CAN Bus Adapter

3.1 Short the Termination Resistor

Toggle R1 and R2 in the blue dip switches of the CAN1 channel on the left side of the CANalyst-II analyzer to the OFF position on the top; and Toggle R1 and R2 in the red dip switches of the CAN2 channel on the right side to the ON position on the bottom.

Because the CAN1 channel is a high-speed CAN and the CAN2 channel is a fault-tolerant CAN, theoretically speaking, the two buses cannot be connected together, and the data sending and receiving of the two channels cannot be realized. In practice, to ensure that HL does not access the terminal resistor, between the two, or

can send and receive data.

3.2 Connect the CANH and CANL Signal Lines of CAN1 and CAN2 Channels.

Short the CANH of CAN1 channel and CANH of CAN2 channel, and short the CANL of CAN1 channel and CANL of CAN2 channel. The schematic diagram after shorting is shown below:

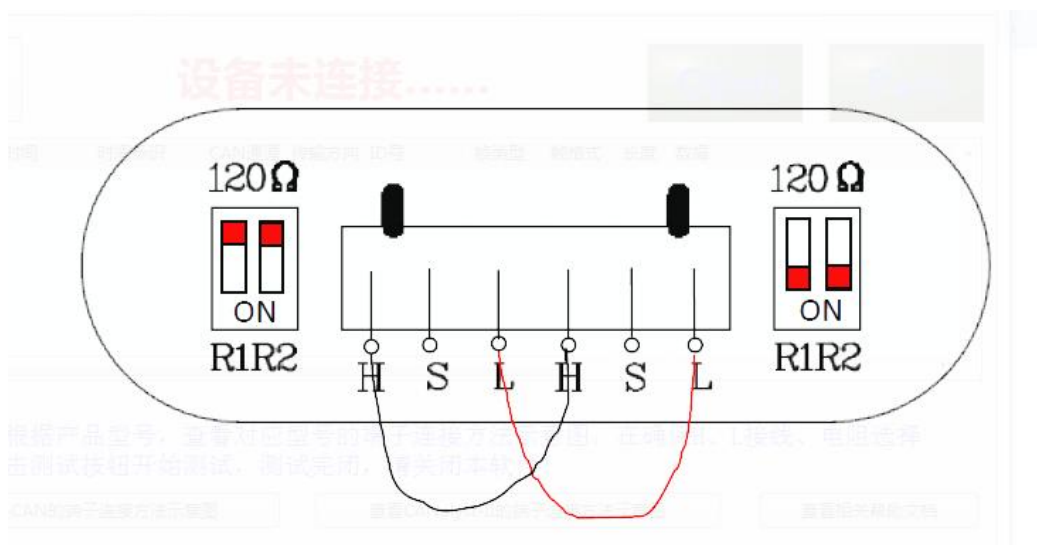


Figure 1 CANalsyt-II Analyzer Test Wiring Diagram

4. Run the USB-CAN Tool Test Tool

Run USB-CAN Tool debugging software, if the computer is installed in the protection software, such as 360 security guards, may open the exe program, a pop-up dialog box prompts the program to modify the registry entries, allow can be, which is LabVIEW run engine in the run of some of the program's operations:



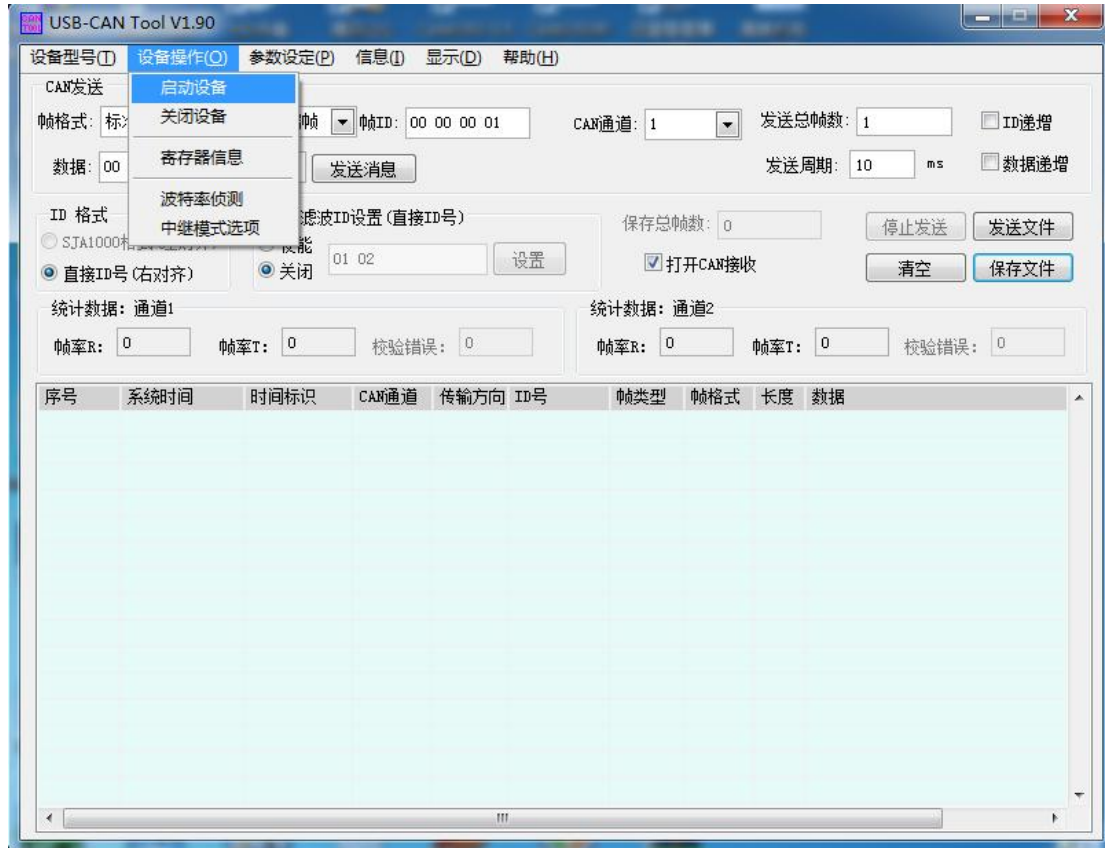
4.1 Automatic Testing With the USBCAN Test Tool

In the USB-CAN Tool interface menu, click "Device Operation USBCAN Test Tool":

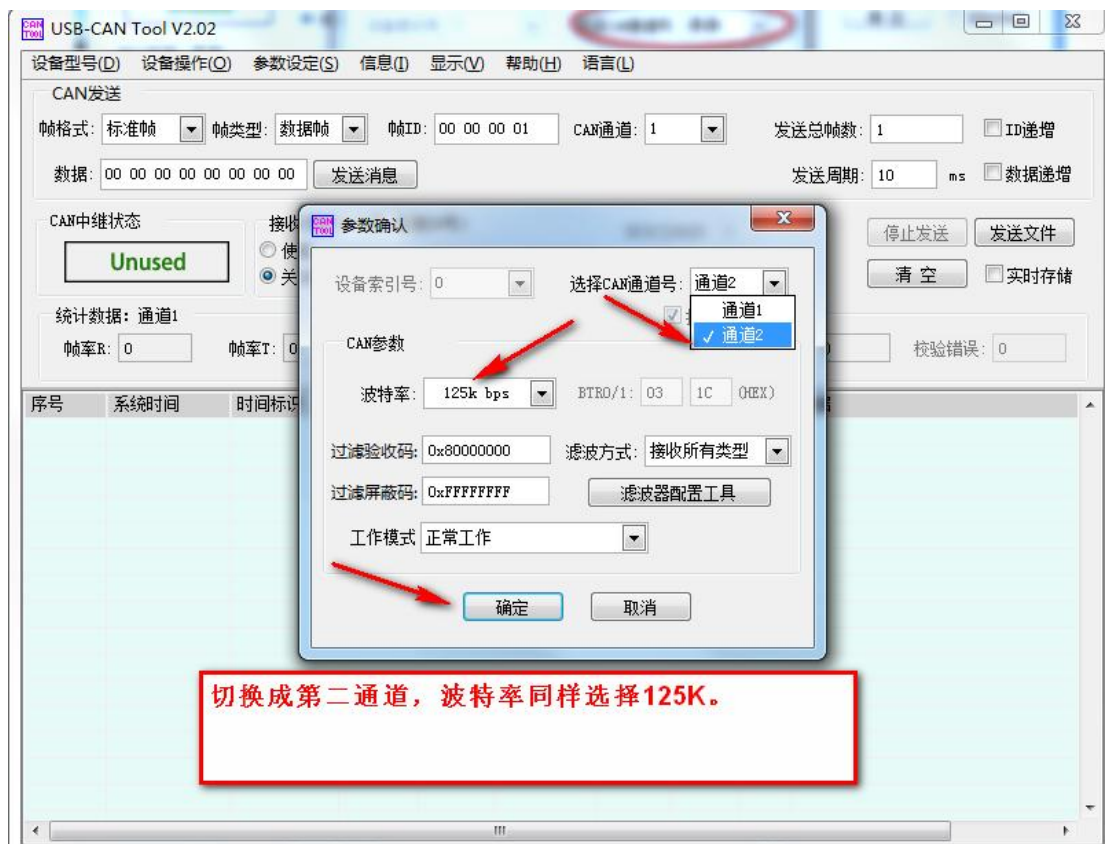
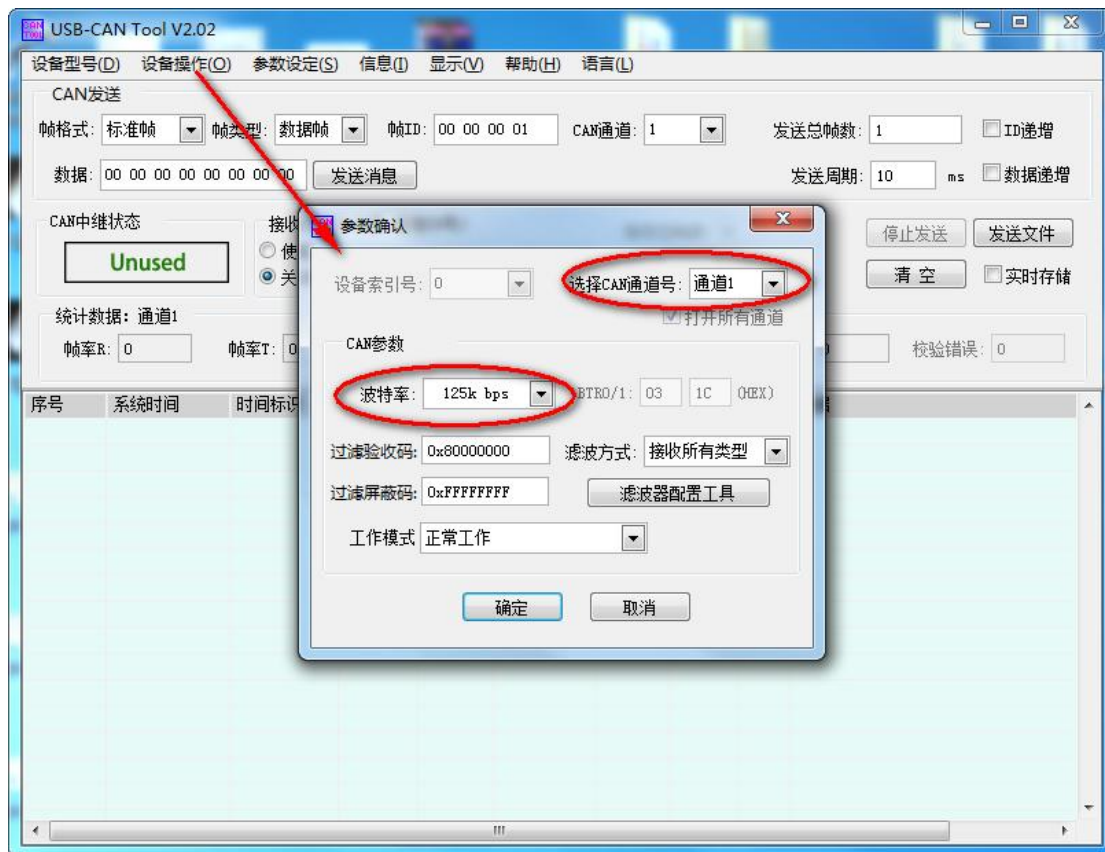
The baud rate of the automatic test tool is 1 M. CAN1 is a high-speed CAN with a maximum baud rate of 1 M; while CAN2 is a fault-tolerant CAN with a maximum baud rate of 125 K, so the test software cannot be used.

4.2 Manual Test

In the pop-up USB-CAN Tool interface menu, click on the "device operation start device":



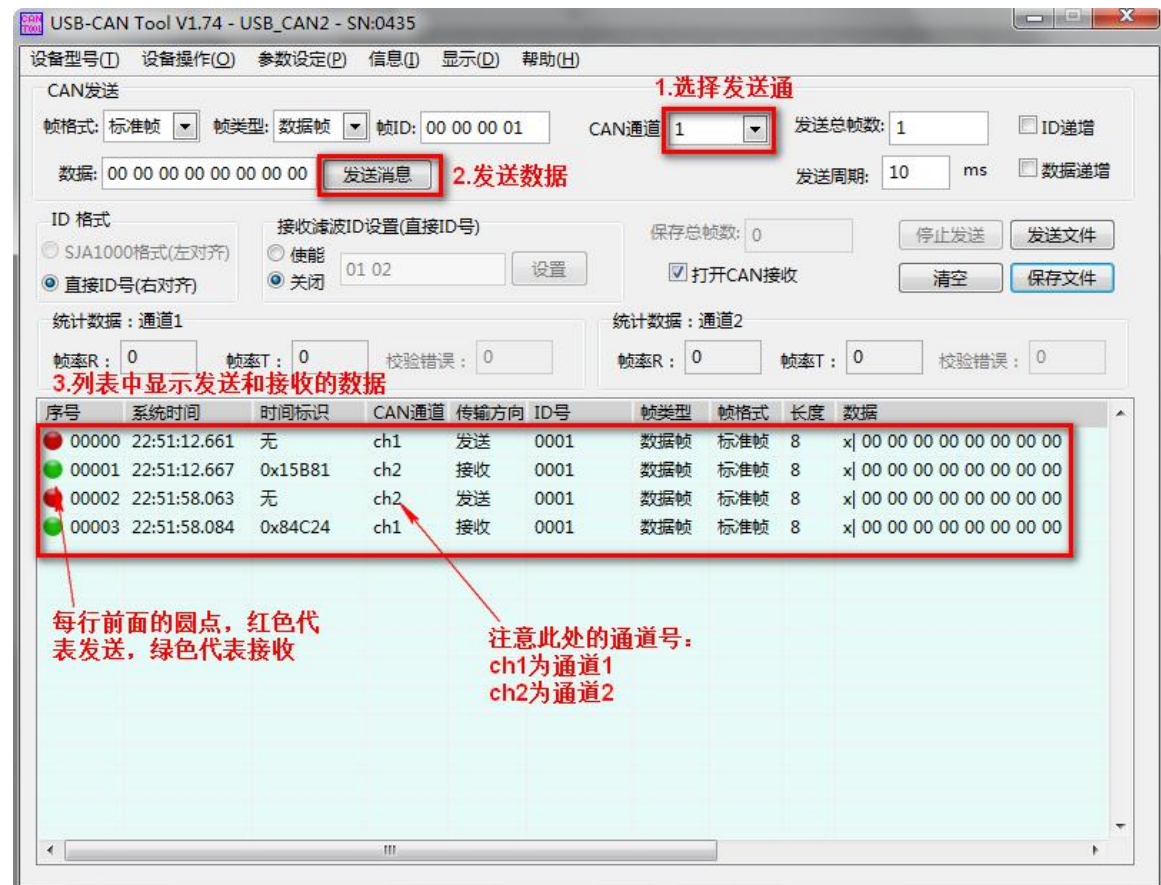
In the next pop-up initialization parameters dialog box, the CAN1 channel and CAN2 channel parameters are set to the same baud rate, where the two usual channels are selected fault-tolerant CAN the highest baud rate of 125 K. Normal mode of operation, in general, the software's default parameters that meet the requirements, click "OK" can be:



After opening successfully, select CAN1 channel on the interface, click "Send" button to send a frame of data, at this time, CAN2 channel will receive a frame of the same data, indicating that CAN1 channel sends normal, CAN2 channel receives normal.

Then switch the sending channel to CAN2 channel, click "Send" button to send a frame of data, at this time, CAN1 channel will receive a frame of the same data, indicating that CAN2 channel sends normally and CAN1 channel receives normally.

Note: If no data is received after clicking "Send", please check whether the terminal wiring is firm!



So far, CAN1 channel and CAN2 channel transmission and reception are normal, it means that the USB-CAN bus adapter is normal.

5. When the Test Is Normal By the Above Method, But It Can Not Send and Receive After Accessing the Bus, It Will Be Processed.

After the above steps are tested normally, it means that there is no problem inside the USB-CAN bus adapter, if it still can't receive or send data normally after connecting to the bus, most of them are caused by the physical wiring and parameter matching errors, please further confirm the following sections:

5.1 Physical Wiring

Strictly follow the above steps for wiring.

Make sure the H and L of the two channels are connected reliably.

Make sure that both dip switches of CAN1 channel are set on the top, and no resistor can be connected.

Ensure that both dip switches of the CAN2 channel are dialed below to access the termination resistor of the fault-tolerant CAN.

In the case of wiring according to the schematic diagram, unplug the USB cable from the computer, and use a multimeter to measure whether the resistance between CANH and CANL is infinity, and infinity means that the bus is normal.

5.2 Parameter Matching Error

Strictly follow the above steps to configure the parameters, if you use it for the first time, directly use the default value. If the configuration has been modified, it is recommended to use the menu "Parameter Setting" -> "Restore Factory Configuration Parameters" and then operate according to the instructions.